Capstone Project 2

General Format

The manuscript must follow the document formatting standards:

- Paper
  - Size : 8.5 inches x 11 inches (short size bond paper)
  - Orientation : Portrait (except for special diagrams)
  - Substance : 20 (for copies to be bounded)
- Spacing : Double
- Line Spacing
  - Before : 0 pt
  - After : 0 pt
- Indention : 0.5 inch (1.27 cm)
- Text Alignment : Left Align (*except for abstract and approval sheet-justified)*
- Margins
  - Top : 1 inch
  - Left : 1.5 inches
  - Bottom : 1 inch
  - Right : 1 inch
  - Gutter : 0
  - Header : 0.8
  - Footer : 0.8
- Font
  - Size : 12 pt
  - Type : Times New Roman
  - Color : Black
- Pagination
  - Upper-right (no extra characters)
  - No page shown on first page of every chapter title page
  - Preliminary pages - use lowercase Roman Numerals (bottom-center)
- Layout for Tables
  - The table caption must be at the top of the table. The caption consists of the table number and the table title (Please see attached sample)
  - The table number must be in Roman Numerals. All table numbers must be in sequence from Chapter 1 until the end of the manuscript.
  - Down below the table number is the table title in Title Case (not italics).
  - The table header has a border on top and bottom only (APA format)

    - The table data cells do not have any borders
    - There must be a bottom border of the table
    - The table must be 3 single spaces below the previous paragraph and 3 single spaces above the next paragraph
- Layout for Figures
    - The image, figures, diagrams must be at the top of the caption
    - The figure must be 3 single spaces below the previous paragraph and 3 single spaces above the next paragraph
    - The figure number must be in Arabic Numbers and must be in sequence all throughout the manuscript.
    - The figure caption must be below the figure. It consists of the figure number and the title (Italicized the title, not the figure number)
- Chapter Title Page
    - Chapter header must be in Title Case and the chapter number must be in Arabic numbers (1, 2, 3…)
    - Chapter title must be in all capital letters
    - Chapter header and Title must be centered and in boldface
- References
    - Citation Style: American Psychological Association (APA) 6th or 7th edition (Hanging Indent – single spacing, double spacing in between paragraph)
- Appendices
        A. Letter to Conduct a Study
        B. Questionnaire
        C. Gantt Chart
        D. Organizational Chart
        E. Researcher's Profile / Curriculum Vitae
        F. User's Manual
        G. Code Excerpts

Others:

1) Align left the Second Level Subheading written in Title Case (ex. Background of the Study, Definition of Terms, etc.)
2) Indent the Third Level Subheading written in sentence case ending with a period.
3) Use the third person point of view throughout the study (Don't use "I", "we", "you", "us", "our", etc.).

The Preliminaries

The preliminary part of a Capstone Project consists of: Title Page, Approval Sheet, Acknowledgment, Abstract, Table of Contents, List of Tables, and List of Figures, appearing in this order. The Title Page, though counted, remains unnumbered. All other pages in the preliminaries are given lower case Roman numerals placed at the center bottom of the page. Capitalize the first letter of the major words in the headings.

Title Page

The section presents the title, the full name of the researcher/s and the submission statement, including the faculty or school, the degrees sought, and the month and year when the degree is to be granted. It should be typed in single space. The title should be concise, descriptive, and comprehensive. It should be expressed in not more than 12 words presenting the variables under investigation and their relationship. The title appears entirely in capital letters. The rest of the title page entries should appear at the center in upper- and lower-case letters.

Book Cover Page

E-PORTFOLIO WEB-BASED SYSTEM FOR SECONDARY SCHOOL TEACHERS

A Capstone Project
Presented to the Faculty of
Iloilo Science and Technology University
Miagao, Iloilo, Philippines

In Partial Fulfillment
of the Requirements for the Degree
Bachelor of Science in Information Technology

by

Name of Leader First

Angel Ann V. Tomesa
Gayo, Ma. Edda Rose A.
Moncal, Givi Ann G.
Mondejar, Christopher G. Jr.
Muyco, Jenny T.
Tidula, Kathy Mae T.
Tolones, Rhamella T.

Month & Year of Graduation

June 2024

Title Page

TITLE OF THE STUDY (Font size: 12, Style: Times New Roman)

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4 Single Spaces

A Capstone Project

2 Single Spaces

Presented to

2 Single Spaces

The Faculty of Computer Studies

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Iloilo Science and Technology University – Miagao Campus

Miagao, Iloilo

4 Single Spaces

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In Partial Fulfillment

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of the Requirements for the Degree

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Bachelor of Science in Information Technology

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4 Single Spaces

by

4 Single Spaces

Full Name of Researcher/s (Firstname MI, Lastname) Leader First
Full Name of Researcher/s (Firstname MI, Lastname)
Full Name of Researcher/s (Firstname MI, Lastname)

4 Single Spaces

Month Year of Graduation

Approval Sheet

The section provides space for the signatures of the Adviser, the Research Committee Members, and the Campus Administrator indicating their acceptance of the research paper.

Approval Sheet

This research entitled "Online Student Grade Access Portal" prepared and submitted by Dexter L. Basco, Paul Bryan G. Gavan, Femma Joy M. Medina, Rodrigean N. Nacis, Khreshia Mae S. Saquine, Jessa Mae S. Siva, and Ela Mae R. Tuvilleja in partial fulfillment of the requirements for the degree of Bachelor of Science in Information Technology, is hereby approved.

JURECA F. NACIONAL, Ph.D.
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Campus Administrator

November 2023

Acknowledgment

The section contains expressions of appreciation for the assistance and guidance extended to the researchers and is limited to two pages.

Acknowledgment

We would like to express our deepest appreciation and gratefulness to all of those who contribute a lot to the accomplishment of this study.

To the Barangay Officials and Concessionaires of Barangay Roma, San Joaquin, Iloilo, for providing us with the opportunity to conduct the study and for all of the resources and support they provided. Their willingness to share their experiences and insights have been invaluable to our study and has helped to make this study a success.

Mr. Dyck Andre S. Robles, for serving as a research coordinator and played a pivotal role in overseeing, ensuring adherence, and conducting timely checks on this paper, contributing significantly to the completion of this study;

Engr. Mario E. Valderrama Jr., our research adviser, whose guidance, recommendations, and invaluable support greatly enriched this project. Additionally, we appreciate his patience, efforts, and the wealth of knowledge he shared with us;

To the ever-supportive parents, relatives, and friends for providing their needs, love, patient, understanding, guidance, encouragement and especially for financial support;

And above all, to the Almighty God who gives His ample blessings and kindness, and who continues to aid and direct the researchers from the start until the end of the study.

Thank you very much and more power!

LONITO B. TACAN, JR.

IRISH JOY T. ECHALAR

JEZZEL T. ESTORNINOS

GEORGIA M. NAQUITA

RONA S. SARSUA

EM SHAIRA F. SANTANDER

RAYMUNDO T. VILLAREN, JR.

Abstract *(Note: It must be 150 – 200 words only)*

The section is a brief and comprehensive summary of the content of the research paper and is limited to one page typed as a single paragraph without indentation and with justified alignment. It includes four or five most important concepts, findings, and implications. Present tense is used to describe conclusions drawn or results with continuing applicability. However, the past tense is used to describe specific variables manipulated or outcomes measured. Verbs are used rather than their noun equivalents. Active form of the verb is used rather than the passive form. Keywords are extracted from the title of the study. Type keywords in italics two spaces from the last line of the body of the abstract.

Abstract

The Online Student Grade Access Portal is a web-based system which aimed to provide services to students, adviser, registrar and administrators of St. Therese's High School, Inc. The system provides a user-friendly interface for collecting and maintaining all types of student data and it has the following features: the efficient way of creating an account for system access, enrolment of students to the system, entering and managing of student's grades and information, access and viewing of student's grades and generating reports. The encoding, posting, and viewing of student's personal information, and subject grades per semester can be accessed and processed online. Feature Driven Development (FDD) methodology was employed in developing the system. The system was developed with the use of Word Press with the programming languages involved CSS, PHP, HTML and MySQL Based on the findings of the study, the design of the system has been responsive to the needs of the students, advisers, and school management in their enrolment and in updating and viewing of grades processes. Furthermore, this system is a solution that eliminates the needs for manual process in storing the data, retrieving and viewing of students' grades..

Keywords: *online, student grade, access portal, web-based system, website*

Table of Contents

The section presents the list of all the sections in the study -- preliminaries, the chapter titles, the main headings and subheadings in the text, reference materials and appendices. The beginning page number of each section is indicated along the right-hand margin. The pagination of the chapters and the wording, capitalization, and punctuation of titles and headings should be exactly the same as they are in the text.

Table of Contents

		Page
Title Page		i
Approval Sheet		ii
Acknowledgment		iii
Abstract		iv
Table of Contents		vi
List Tables		vii
List of Figures		vii
List of Appendices		ix
Chapter		Page
1	INTRODUCTION	1
	Background of the Study	1
	Objectives of the Study	3
	Conceptual Framework	4
	Definition of Terms	5
	Significance of the Study	7
	Scope and Limitation of the Study	8
2	REVIEW OF RELATED LITERATURE	10
	Related System	10
	Developmental Tools	11
	Accessing Student Portal	13
	Online Test Web Portal	14

List of Figures

The section presents the figure titles and the corresponding pages where they appear in the body of the paper. Figure titles in this section should not be in italics.

List of Figures

Figure		Page
1	The Conceptual Framework of the Study	4
2	System Development Methodology	18
3	Use Case Diagram of the Student	22
4	Use Case Diagram of the Adviser	23
5	Use Case Diagram of the Administrator	24
6	Use Case Diagram of the Registrar	25
7	Activity Diagram for the Student	26
8	Activity Diagram for the Adviser	27
9	Activity Diagram for the Administrator	28
10	Activity Diagram for the Registrar	29
11	Network Architecture	32
12	Log in Screen for Student	33
13	Home Page	34
14	Creating an Account for User	35
15	Enrolment of the Student to the System	35
16	Students' System Enrolment	37
17	Advisers' Interface	38
18	Students' Grade Interface	39
19	Administrator Dashboard Interface	40
20	Students' Grades View Interface	41
21	Students' Grades Report Interface	42

List of Tables

The section should follow the Table of Contents. It presents the table titles and the corresponding pages where they appear in the body of the paper. Table titles in this section should not be in italics.

List of Tables

Table		Page
I	Hardware Specification	30
II	Software Specification	30
III	Network Specification	31

List of Appendices

List of Appendices

Appendix		Page
A	Letter to the Conduct a Study	52
B	Questionnaire	53
C	Gantt Chart	55
D	Organizational Chart	56

The Research Chapters

Specific Guidelines for every chapter:

Chapter 1:

Contents:

1. **Background of the Study**
2. **Objectives of the Study**
3. **Conceptual Framework**
4. **Definition of Terms**
5. **Significance of the Study**
6. **Scope and Limitations of the Study**

Background of the Study

The background of the study is a part of the research provided in the introductionsection of the paper. It is the compilation of adequate information based on the analysis of the problem or proposed argument, steps and methods needed to arrive at the design and the implementation of the results achieved and feasible solution (Lopez, 2019).

According to Santos (2015) in writing the background of the study begin by giving a general overview of your thesis topics and introduce the main ideas you will bemaking use of throughout your thesis. Cite some sources of your research to avoid plagiarism.

Present the problems or observations encountered in the current process and discuss your solution why you need to develop a system. How will it solve the problem.

***Note**: 1st paragraph – Introduction; 2nd paragraph – readings or articles that support the introduction; 3rd paragraph – discuss the present problems; 4th paragraph – discuss the developed system as solution to the identified problems.*

Example:

Chapter 1

INTRODUCTION

Background of the Study

Information Technology (IT) support for end-users has emerged as one of the leading concerns of organizations. Continuous adapting and updating of new technologies have made development of effective and efficient help desk services challenging for organizations. In today's trend, organizations have actively searched fornew ways to provide better help desk services that can satisfy their growing customer demands and expectations (Gonzales & Santos, 2015).

These days, almost all government agencies, academic and medical institutions and other private companies have their own IT departments. IT staffs or Technical Support Staff are in-charge of handling queries and resolutions to IT-related issues (Bayona 2061). One of the government agencies is the Department of Public Works and Highways that undertakes the planning, design, construction and maintenance of all infrastructure projects in coordination with concerned stakeholders. The department provides its Regional and District Offices of Information and Communications Technology (ICT) resources for employees to effectively perform their duties and responsibilities. The District Information Technology (IT) Support Officers troubleshootsand helps users in the resolution of ICT problems to ensure that

workstations and other peripherals are operational and that downtimes if unavoidable, are reduced to minimum; provides support for application deployment and its use; and track ICT problems and recommend enhancements.

During the conduct of this research, IT Support Officers were interviewed. Based on the data gathered, common reported issues of employees are printer problems (paper jammed, unable to print, cartridge replacement), computer works slowly or hangs up, unable to power on computers, and application or system errors (forgot password, runtime error, invalid username/password). In the current setup, these problems are received through phone calls, emails and walk-ins mostly at the same time of the same day. These issues need to be prioritized on which one needs urgency or immediate response to reduce downtimes at the workplace and streamline communication to providequick resolution for immediate needs and maximize IT service functions (Inquirer.net).

With the aim to improve the technical support given to employees, the researchers were motivated to develop a helpdesk management system that keeps track of employees' service requests, resolves issues, maintains a record of employees' assets, and manages list of reports needed by the district office. The system can facilitate the communication between employees and the office to improve the reliability and usability in handling helpdesk management. It improves the quality of assistance and support provided to the employees and allow technical support staff to save crucial time and efforts in handling customer or employees' complaints and requests, thus, minimizes the total time spent on resolving a problem.

(Describe what objective is)

Objectives of the Study

The main objective of this study is to develop a Help Desk Management System using Priority-Based Scheduling Algorithm of SEAFDEC, Tigbauan, Iloilo.

Specifically, this study aimed to:

a. create a ticket automation and queuing for requests, problems andinquiries;
b. create a scheduling scheme of help desk tickets based on priority;
c. design a knowledge-based system for common help desk issues; and
e. generate and manage report.

(Describe what Conceptual Framework is)

Conceptual Framework

The conceptual framework of the study is illustrated in Figure 1 showing theinput, process and output.

Figure 1 illustrated the conceptual framework of the study.

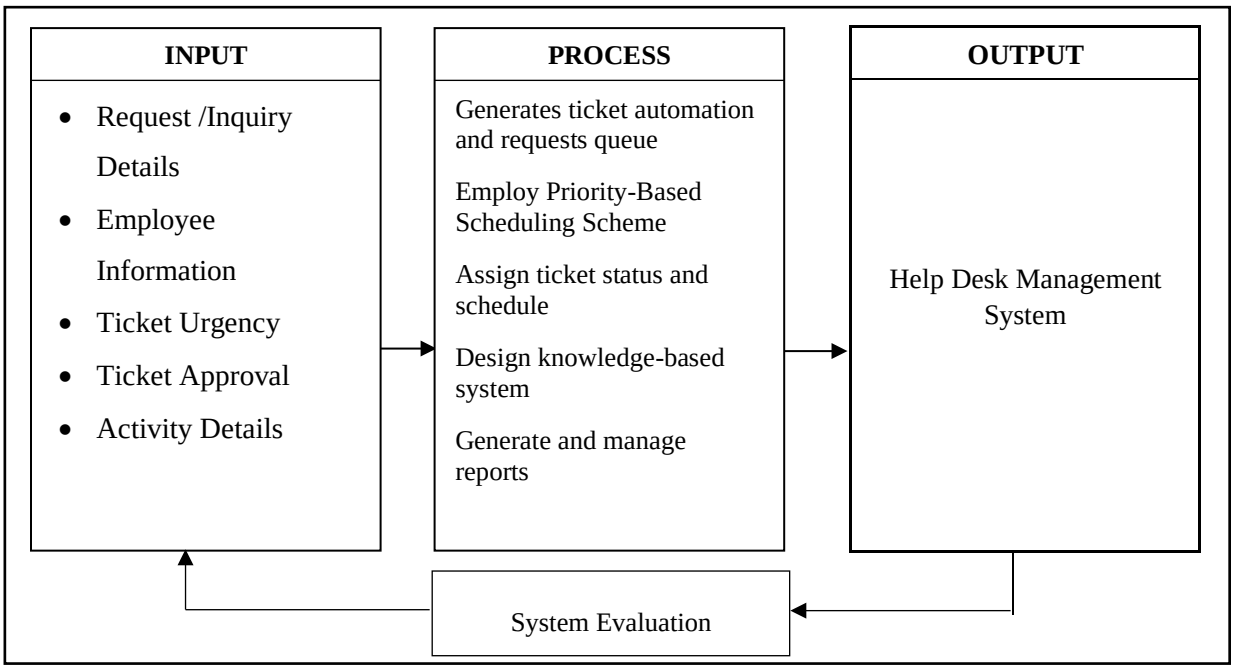


Figure 1. *The Conceptual Framework of the Study*

The input data will come from the end-users, these are the employee informationand request/inquiry. These data contain the information logs of service request, complaints and queries, the date of request, problem or request description, product name, and office assignment.

Another group of users are Agent which refers to the IT Staff, then the Supervisorwho functions as the Unit Head or the Head of section where the request

came from and the Administrator acts as the IT Support Officer which provides technical support to end-users and has the full control in using the system. The input data contains the ticket urgency (low, medium and high), ticket approval and activity details which refer to the action taken performed in addressing the problem or queries by the end users.

The processing involves the generation of ticket automation and requests queue coming from the different requests. A priority-based scheduling scheme is employed to manage the assignment of ticket status and schedule and of request. A knowledge-based system was designed to store all the processing methods and procedures for resolving theissues. Lastly, it was also designed to generate and manage reports for documentation purposes.

The output of the system is a help desk management which is to be evaluatedusing the ISO 25010 criteria for feedbacking mechanism.

(Describe what Definition of Terms is)

Definition of Terms

The following terms were defined conceptually and operationally for betterunderstanding of the study:

Agent. A person who works with customer requests (dictionary.com).

In this study, an agent refers to the Unit Heads, and Supervisors who process theticket approval.

Employee Information. It is a key data on employees that can be used to keeptrack of who worked for the company, when, and in what positions (Reyes, 2018).

In this study, employee information refers to the employees ID, name, position,and office assignment.

End-User. A person who interacts with a system, typically through an interface, toextract some functional benefit (cisco.com).

In this study, end-users refer to all DPWH employees who were granted access to use the ICT resources.

Help Desk Management System. A computer program that enables customer-careoperators keep track of user requests and deal with other customer-care-related issues (eticket.net).

In this study, help desk management system refers to the developed system for DPWH which cater request for ICT related problems.

(Define What Significance of the Study Is)

Significance of the Study

The development of Help Desk Management would provide significant benefits to the following:

IT Support Officer. The District IT Support Officer of the DPWH-Antique wouldefficiently monitor and track queries, request and complaints. There would be a single point of communication between employees without any hustle. It allows to improve thequality of service and the technical support provided to employees through systematic approach.

Employees. Employees from different offices and sections would have satisfactionand maintains profitable relationship with the IT Support Officer. Waiting time is reduced and reported issues or queries are fixed quickly and promptly.

Offices/Sections. Different offices/sections would have good service. This allows catering of more complaints and service requests in a day or in a short period of time and boost employees' productivity. Each section/office would have more outputs accomplish with less downtimes at work.

Department of Public Works and Highways (DPWH). The ICT resources used by employees are likely to be more operational at all times with minimal interruptions. Thus,workforce and workflows are optimized providing better service to the public. This will improve the performance of agency in delivering quality works and services.

Researcher. The success of this study will help improve the performance of everyDistrict IT Support Officers in giving technical support services to end-users. This will serve as a tool in handling and managing their job description effectively.

Future Researchers. This study will serve as a model and basis of future development with scalable and complex features.

(Describe what Scope and Limitations are)

Scope and Limitations of the Study

This study was conducted at the Department of Public Works and Highways (DPWH) Antique District Engineering Office. All the data were gathered and collected from the district office. The help desk management focuses only on all IT-related issues,requests and inquiries.

The system is designed to run on workstations compatible to a required specifications and connected to a local area network, online features are not included inthe development of the system. The developed system contains the following features: help desk management, IT asset inventory, knowledge-based solutions and report management.

Incident escalation is only between the employees, the agent and the IT SupportOfficer and the system can be utilized in either offsite or onsite assistance. For onsite, generated tickets are given scheduled date and time of resolution. Either or both of the Agent and IT Support Officer conduct an onsite inspection to know the nature of the problem and its urgency of use. For offsite use, the employee can make use of the knowledge-based system solutions which manages common and recurring technical issues and problems. Pending requests will not be processed during holidays, weekends, and beyondoffice hours. Instead, it will be scheduled on the next office hour.

Chapter 2:

The Research Services Division (RSD) Research Manual [1] specifies that in writing a Review of Related Literature, the researcher must discuss the following:

1. **Thematic/Topical** approach which presents the concept, idea, observation, belief, methodology, etc. used in the study.

2. **Existing studies/prior arts** which presents related or existing studies, and designs related to the study being conducted.

3. **Synthesis** which connects the concepts and prior art to the study and addresses the gap or problem one is trying to address.

Chapter 2

REVIEW OF RELATED LITERATURE

This chapter provides the idea on how these studies are related to the present research. Theoretical and technical definitions related to this study are further discussedto promote understanding and cycle of what a help desk entails.

Help Desk Management

Moussas (2015) stated that a help desk is a multi-dimensional resource, designated to help in reducing downtime in IT services and functions and making them available for maximum time. It is specially focused on end user functionality, and, thus,is responsible for quick resolution of immediate needs, incidents and technical issues ofend users.

Some of the features commonly found in a help desk software include convertingemail to ticket; segregating incidents on the right category and sub-category; assigning totechnician; setting up SLAs based on criteria; pulling out knowledge-based

solutions filtered by category; automating help desk workflow; providing different access privileges; and analyzing help desk performance.

The idea on help desk management is the main concept anchored in this study to develop a system that tracks IT related issues and give technical support to employees in utilizing ICT resources used in their workplace to improve their productivity. All issues and inquiries through telephone calls or walk-ins are converted to tickets, categorized andassigned with priorities which are some of the help desk processes included in this study. An inclusion of the knowledge-based repository of frequently recurring issues is also adapted in this study to provide employees a procedural guide on the use and maintenance of some ICT resources.

Republic Act No. (RA) 9485

In government issuance of the “Republic Act No. 9485”, is also known as the "Anti-Red Tape Act of 2007" or An Act to Improve Efficiency in the Delivery of Government Service to the Public by Reducing Bureaucratic Red Tape, Preventing Graftand Corruption, and Providing Penalties Therefor [19]. It is stated in Section 8 - Accessing Frontline Services, that:

(1) Acceptance of Applications and Request –

a. All officers or employees shall accept written applications, requests, and/or documents being submitted by clients of the office or agencies; (2) The responsible officer or employee shall acknowledge receipt of such application and/or request by writing or printing clearly thereon his/her name, the unit where he/she is connected with, and the time and date of receipt; and (3) The receiving officer

or employee shall perform a preliminary assessment of the request so as topromote a more expeditious action on requests.

(2) Action of Offices –

All applications and/or requests submitted shall be acted upon by the assigned officer or employee during the period stated in the Citizen's Charter which shall not be longer than five working days in the case of simple transactions and ten (10) working days in the case of complex transactions from the date the request or application was received. Depending on the nature of the frontline services requested or the mandate of the office or agency under unusual circumstances, the maximum time prescribed above may be extended. For the extension due to the nature of frontline services or the mandate of the office or agency concerned the period for the delivery of frontline services shall be indicated in the Citizen's Charter. The office or agency concerned shall notify the requesting party in writing of the reason for the extension and the final date of release for the extension and the final date of release of the frontline service/s requested.

Some incident management tools perform automatic calculations for Urgency based on Impact, (Service Level Agreement) SLA and (Operational Level Agreement)OLA involved etc. Most consultants recommend the simple matrix which will automatically calculate incident priority out of the simple value of Impact x Urgency. Recommended granulation of Priority is 4 to 5 different values. Usually the lower the value - the higher the Priority, thus Priority=1 is the highest one, and

Priority=5 the lowest (Chavez, 2012). The description for each Priority depends on the context of theorganization, and on the criteria that needs to consider when the time comes to the Service Desk Agent to establish an Incident Priority. A resolution target will be set foreach Priority; the objective is to resolve incidents within this delay.

Figure 2 illustrates the standard priority matrix.

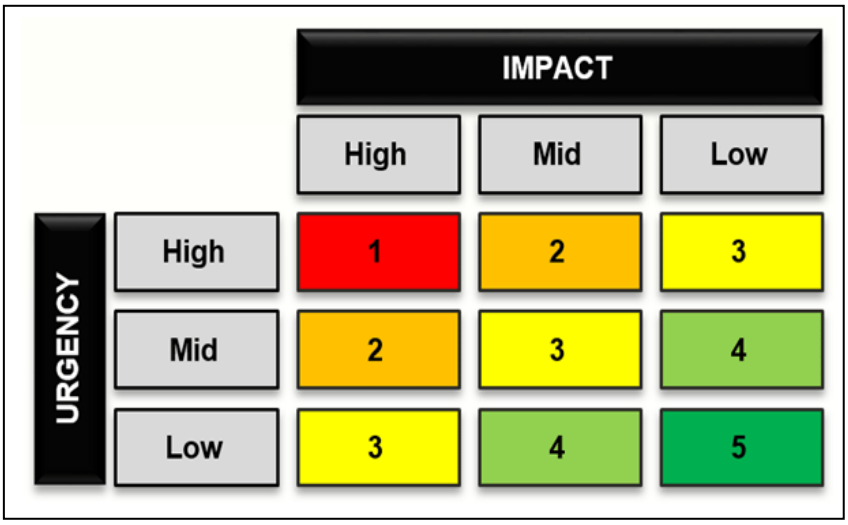


Figure 2. *Standard Priority Matrix*

This study applies the same Standard Priority Matrix to define priorities on each help desk tickets and on how to address each queued issues of the employees. This study established the same with standard priority matrix wherein it is based on impact and urgency of the issue or the problem. The same values were used with 5 as the lowest and1 is the highest priority.

IT Asset Management

IT asset management involves the management of IT infrastructure like computer,monitor, keyboard and even software installers. IT asset management helps in tracking and configuring all IT devices within one information system. It also includes the detailedanalysis of all the assets of the organization (iteara.com).

There are five stages to an asset life cycle: requisition, procurement, deployment,maintenance and retirement. The life cycle actually commences when an asset is requisitioned. Similarly, life cycle management continues after an asset is no longer operationally active, extending to retirement and disposal activities. Management of the life cycle may further continue for an indefinite period where historical asset records must be retained to satisfy legal and fiscal requirements (NewsToday.com).

There are four core capabilities tools that must be provided to support an effectiveIT asset management program such as IT asset management repository, hardware and software inventory and usage information, integration with adjacent IT and business systems, and data import. In evaluating an IT asset, the preferred method of evaluation is through first hand use of fully functional trial or evaluation versions. This provides an opportunity to rigorously test software with real data and fully explore integration with existing systems and environments (Daza, 2015).

This study applies the asset life cycle in managing IT Assets as a part of the help desk management system. Just like the previous study, it supports IT asset management repository; hardware and software inventory; and usage information. In addition, records of employees are associated for each record kept as IT assets for easy access and keeping

track of employees' accountability.

Help Desk System with Intelligent Interface

This study develops a help desk information retrieval mechanism suitable for a wide range of users and to provide a way of easily maintaining the system. The model is developed for use over the World Wide Web combines keyword search and case-based reasoning to provide both rapid focusing on the part of the help information and guided interaction when the user is unclear about appropriate keywords. To develop better methods of maintaining knowledge bases for help desk system, knowledge-based systemsare roughly classified into two groups: rule base and case base (B. H. Kang et al., 2011) . Rules are the most popular representation for the knowledge base in the expert system.

The methods to obtain rules are classified into two categories, automated methods (machine learning) and manual methods. Regardless of which method is used, the rule- based approach constructs a knowledge base which interprets the problem and suggests solutions. Case-based reasoning (CBR) approach is frequently used to build help desk systems. It is based on the cognitive assumptions that real expertise comes from the experience of the expert, and that episodic memory. CBR is appropriate when there is noformalized knowledge in the domain or where it is difficult for the expert to express their expertise in the rule format.

In this study, rule-based approach is adapted to be associated in the knowledge base system module. This study provides scenario of common IT-related issues with supporting procedures on how to solve them. The use of rule-based approach is also applied in this study for setting priority levels on help desk tickets. Each priority level

hasa defined rule or criteria to consider as a guide to classify the reported help desk tickets.

Chapter 3:

Contents:

- Project Description
- Development Process
- Requirement Analysis
  - Use Case Diagram
  - Activity Diagrams
  - Class Diagram
  - Sequence Diagrams
- Design Specification
  - Hardware and Software Requirements

Example:

Chapter 3

METHODOLOGY

This chapter discusses and illustrates the methods and procedures on how thestudy was designed and analyzed to achieve the objectives.

Project Description

The Help Desk Management System using Priority-Based Scheduling Algorithmbasically consists of a help desk for employees and IT Support Officer, inventory of IT assets and knowledge-based problem solutions. The system is an intranet-based technology which is a single point of communication for both the

employees and IT Support Officer. The help desk is used by the IT Support Officer in receiving tickets, assigning priorities, scheduling of on-site inspection for resolving issues and generation of periodic reports. Help desk issues are categorized as to hardware, software, business applications and services. The IT Support Officer can accomplish its inventory on IT assets while handling help desk tickets. Each IT equipment with reported problems can be viewed with its detailed information for keeping track its status and history logs.

Ticket generation of problems and inquiries, viewing of ticket status for any changes andperforming self-support solutions to common IT-related issues are accessible on Employees Dashboard. It provides basic troubleshooting guides to common problems or issues with the application of knowledge-based scheme. A Priority-Based Scheduling Algorithm was employed to manage a fair and balance processing of ticket request.

Ultimately, the system applies metrics to manage help desk tickets through priority. Priorities set for each ticket is assigned from critical, high, medium to low. The system also keeps history logs for recurring tickets and with repetitive reported issues onparticular IT equipment used by the employees.

(Describe project development process)

Project Development Process

Project Development

System Development Methodology refers to the framework and context followedin an organization to plan, design, and controls the process in developing information systems.

Stair and Reynolds, (2010) discussed that the Object-Oriented Systems Development (OOSD) is an approach that combines the logic of systems development life cycle with the power of object-oriented modeling and programming. OOSD follows adefined systems development life cycle, much like the SDLC. The life cycle phases are usually completed with many iterations. The OOSD software development process has three phases: analysis, design and implementation phase. System development can be viewed as a process. Each Process can be divided into small, interacting phases into sub processes which consist of a description of the work to be done, specification of input required for the process, and specification of output to be produced. Software development process also can be viewed as a series of transformations to promote building high quality software (Thomas, 2011).

Figure 3 shows the phases in Object-Oriented Systems Development Life Cyclewhich has three sequences of transformations.

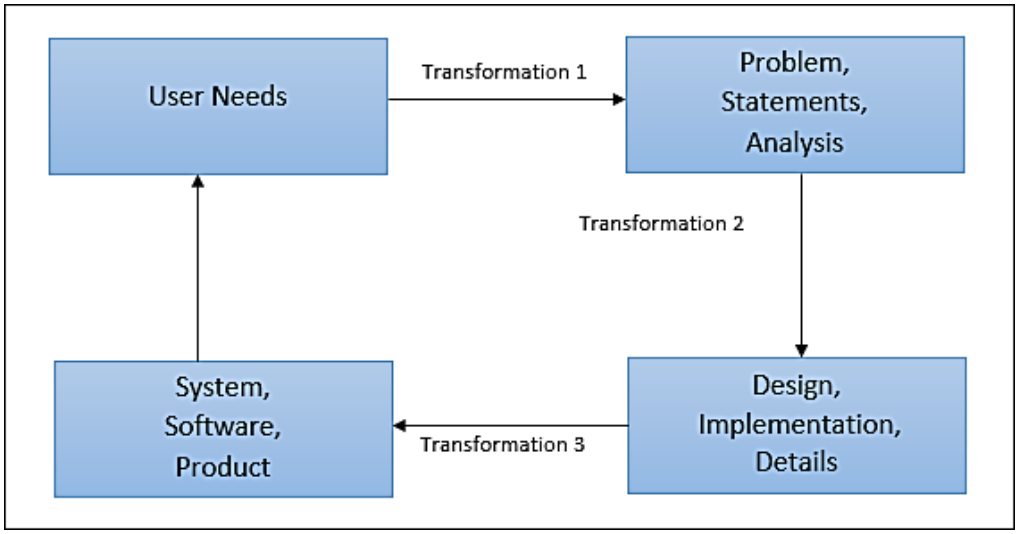


Figure 3. *Object-Oriented Systems Development Life Cycle*

Analysis. In this phase, it specifies the functional behavior of the systemindependent of implementation details.

In this phase, identification of features is performed, and the objectives and scopethat needs to be incorporated in developing the system are set. Activities involve identifying and conceptualizing the current help desk issues and determining system input through monthly help desk and IT inventory reports.

Transformation 1. It translates the user's needs into system requirements andresponsibilities. In this, the system can provide insight into the user's requirements.

In this study, activities are performed such as gathering of important informationthrough interviews with employees and other IT Support Officer, collecting of archives of DPWH published guidelines, and issuance of memorandum and department orders related to the use and management of IT assets.

Design. This phase analysis model is refined and adapted to the environment.During this phase, additional objects and classes that support implementation of the requirements are identified and defined.

In this phase, identifying department's requirements like who will use the systemand how they will use it is being considered. Also, the users of the system were identified, employees and the IT Support Officer; identifying on how employees submit requests and other issues with the current system; and how IT Support Officer provides solutions to these problems encountered. Both users need to register or enroll to access the system.

Transformation 2. It begins with a problem statement and ends with a detailed design. It can be transformed into an operational system. This transformation includes thebulk of the software development activity which also includes the definition of how to build the software, its development, testing, and design descriptions, and the program & the testing materials.

In this phase, the screen lay-out of the system is anchored to the existing job sheetused by the employees in all of IT related issues. The help desk management process andflow of transaction is reviewed and designed to meet the user's needs such as how to assign a help desk ticket and assigning priority with the use of priority matrix and creating sets of rules and criteria. In resolving help desk tickets, offsite and onsite resolutions for users are considered.

Implementation Phase. Design is implemented using a programming language or database management system. During this phase, each feature of the system is coded and developed individuallyand tests its functionality based on the role of users in using the system.

The developmentof different modules: the help desk management, it asset inventory, knowledge-based solutions and report generation are designed and build adapted from the current process flow of the users. Its user interface and database together with the integration of the different functions of the modules is implemented.

Transformation 3. It refines the detailed design into the system deployment. It satisfies the user's needs. It includes the equipment, procedures, and people. It representsembedding the software product within its operational environment.

In this phase, implementation of functionalities on the help desk management, ITinventory, knowledge-based solutions and report generation are reviewed as to its designand evaluated if it meets the user requirements. Each module is also tested upon the development of codes to test for errors.

System Flowchart utilizing the Priority-Based Algorithm

Figure 4 illustrates flowchart of the priority-based scheduling.

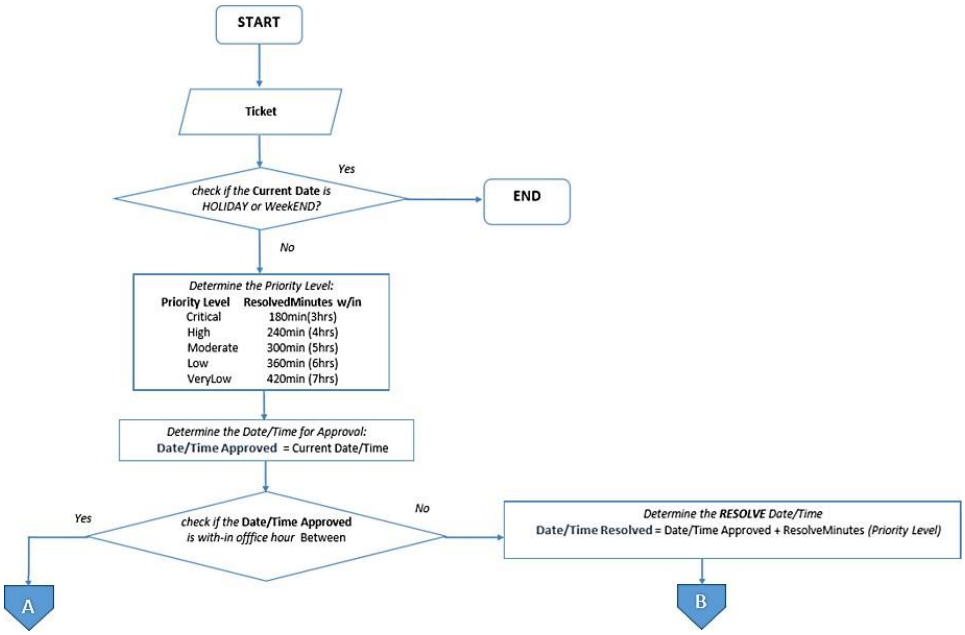


Figure 4. *Priority-Based Algorithm Flowchart*

In the utilization of the priority-based algorithm, the process starts in creating a ticket which checks first if the system's date falls on a holiday or weekend. If it does, then the ticket creation is terminated. Otherwise, the ticket is created and approved. Then,a priority level is assigned, namely: critical as the highest and very low as the lowest priority wherein each priority level has equivalent resolution time to serve the ticket request. Once the priority level is determined, the date and time approved is checked if it is equivalent to the current date and time of the system.

If the approval time is not within the office hours, it is then verified if the time is before or after. Thus, if it is checked before the office hours, then the date and time approved must be equivalent to the current date and beginning time of office hour (8:00)in the morning. Then, the resolution time is derived through: Resolution time = *Date andTime Approved + Resolve Minutes (Priority Level).* Thus, the date and time of approval and resolved date and time is processed and updated in the system. Approval time also means response time wherein resolution time is the same terminology as the burst time.

Then, if the approval time is within the office hours, the system checks if the following day falls on a weekend. If yes, time extension is applied and approval time is updated using: Next Day = *Current Date + 2days (Saturday & Sunday) and beginning office hour (8:00)* in the morning. Else, the system checks if following day is a Holiday,following the approval time is equivalent to the previous computation of the Next Day. The approval and resolved date and time are updated.

If the approval time is within the office hours, the system verifies if the

resolutiondate and time is less than the end of office hour (5:00) in the afternoon, then automatically the approval and resolution time is updated in the system. Otherwise, the process on the time difference of resolution time from the end of office hour is calculatedand added to the computation of the Next day to get the updated resolution time. The same process follows if it falls on weekend or holiday.

To execute the queuing process:

Turn Around Time (TAT) = *Resolve Date/Time – Current Date/Time*

This is to reduce the value of turnaround time while the clock is running. Therefore the queuing process is based on the turnaround time. Wherein, the waiting time is display in the system to show the amount of time spent while on the queue using:

Waiting Time = *Current Date/Time + Resolve Date/Time*

Use Case Diagram

M. Seidl et al., [41] specified that use case diagram allows the description of the possible usage scenarios (use cases) that a system is developed for. It expresses what thesystem should do but does not address any realization details such as data structures, algorithms, etc.

Figure 5 illustrates the Use Case Diagram for User.

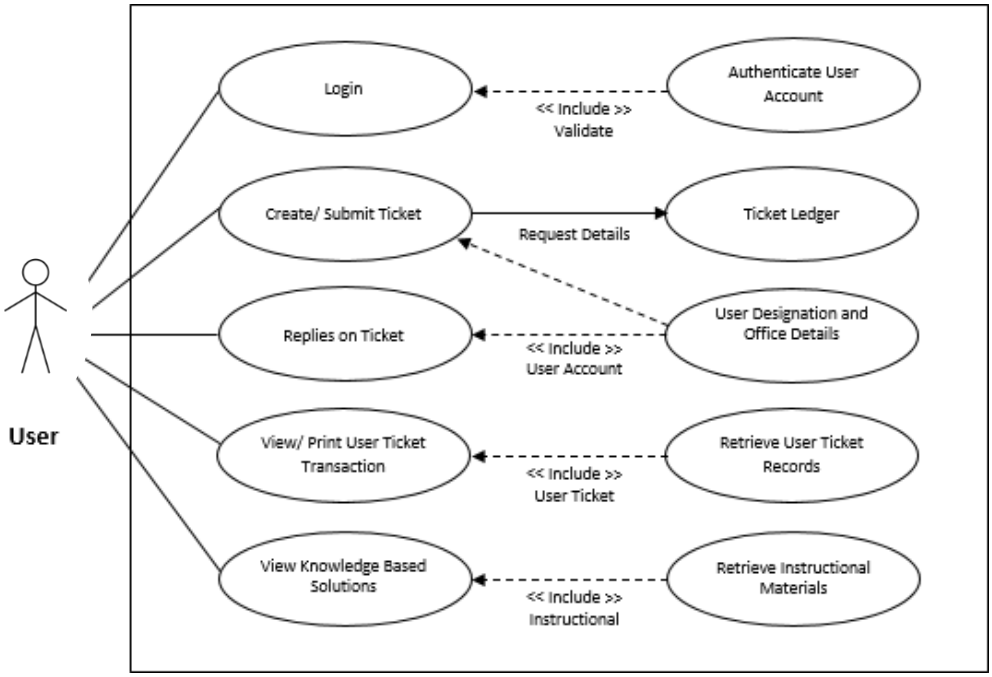


Figure 5. *System Use Case Diagram for User (Employee)*

As shown in the figure, the user of the system creates and submits the ticket, addscomments on the Ledger, views and receives ticket status updates, downloads and prints PDF file of Knowledge Base Solutions available on the system.

Figure 6 illustrates the Use Case Diagram for Agent.

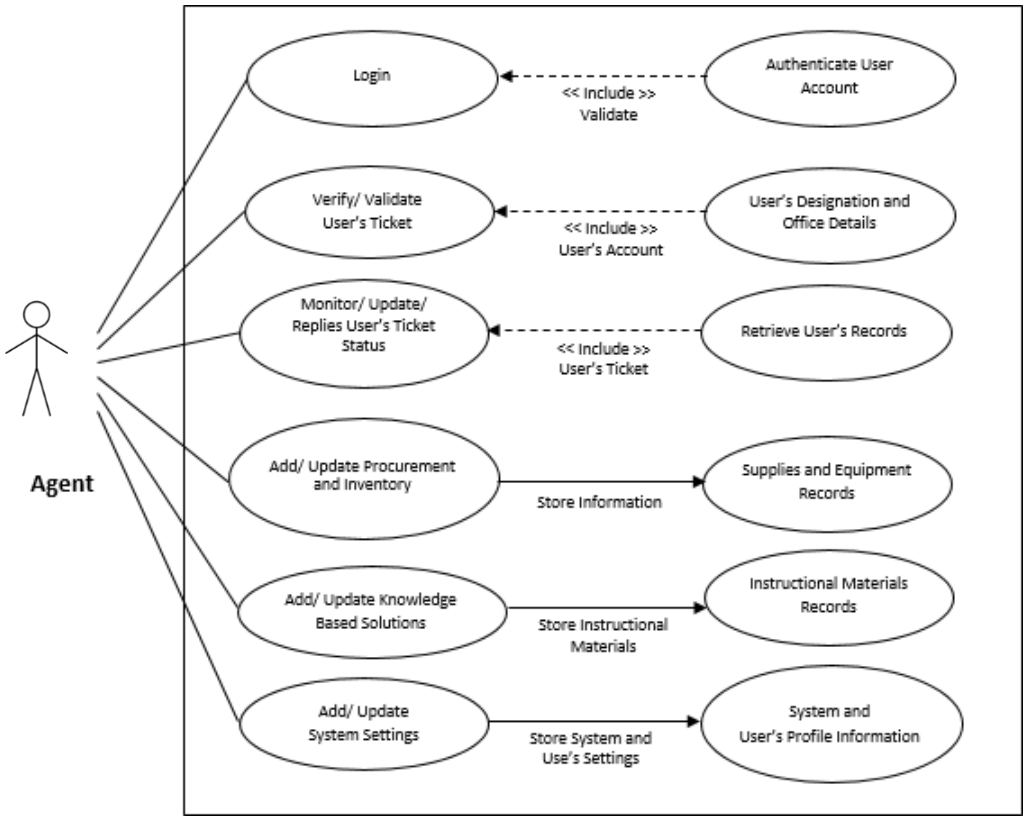


Figure 6. *System Use Case Diagram for Agent (ICT Staff)*

In this process, the Agent is managing the help desk system in terms ofmonitoring and updating the user's ticket status as well as providing proper solutions. The agent is also tasked to generate and upload knowledge-based solutions forinstructional materials, inventory details, system settings, which include setting of holidays and extension and user's profile information. The agent can generate a report which includes the ticket summary of transactions and inventory list of IT equipment.

Figure 7 illustrates the Use Case Diagram for Supervisor

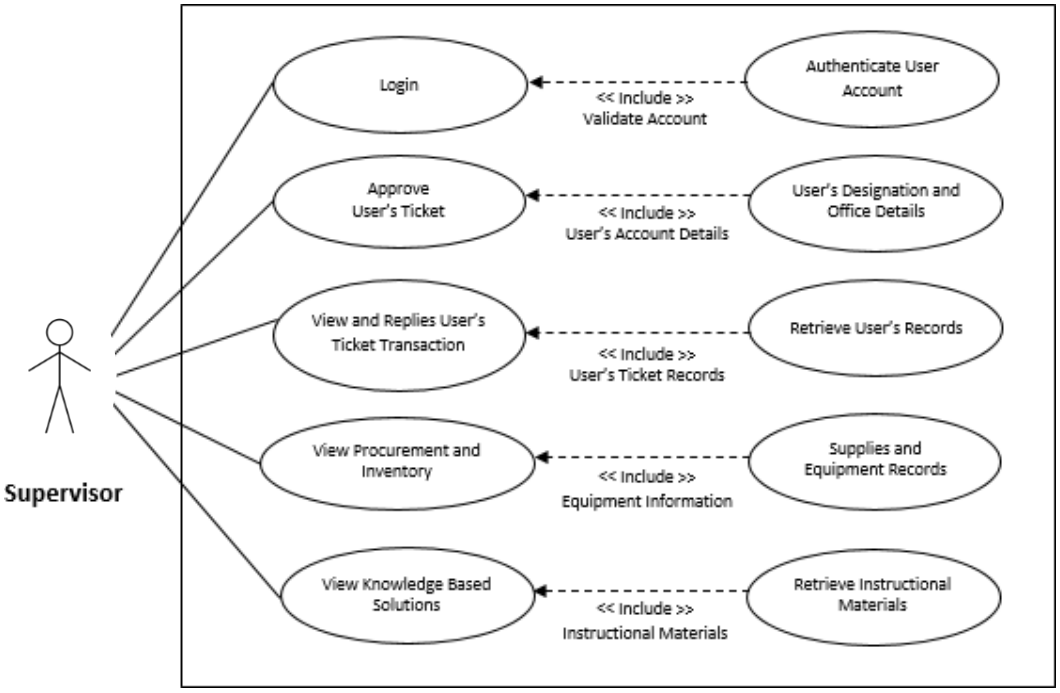


Figure 7. *System Use Case Diagram for Supervisor (Unit Head)*

In this process, the supervisor approves the submitted help desk tickets, adds andupdates comments on the Ledger, accesses list of procurement and inventory, and viewsavailable knowledge-based solutions. The supervisor is also allowed to print or generatereports such as printing of ticket transaction based on the date range given and printing the list of IT inventory.

Figure 8 illustrates the Use Case Diagram for Administrator.

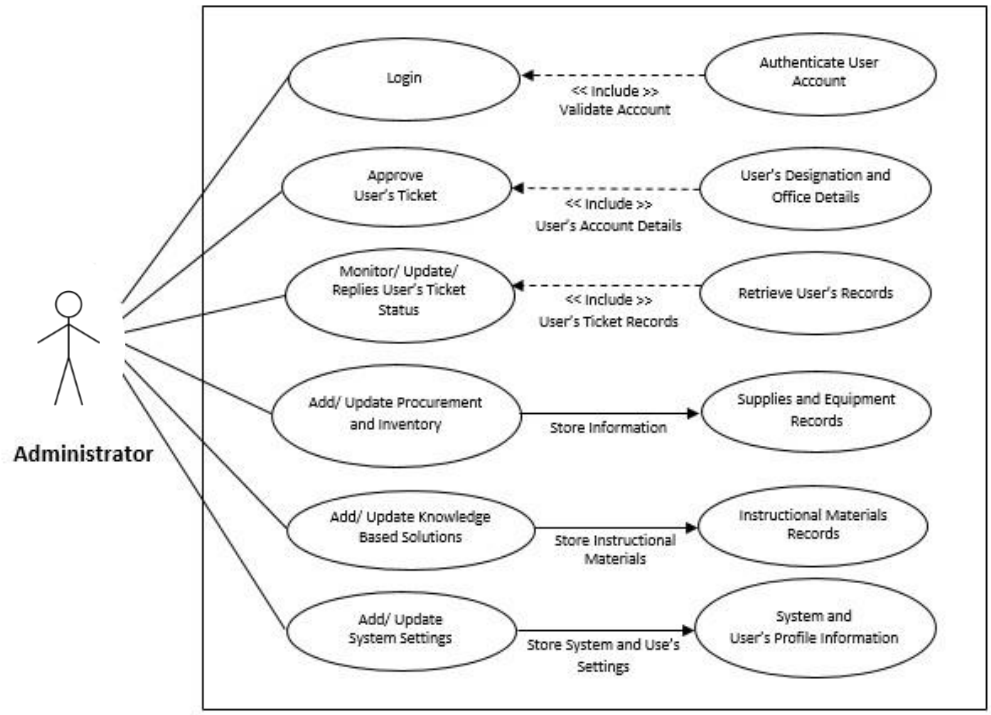


Figure 8. *System Use Case Diagram for Administrator (IT Support Officer)*

For the System's Administration, the Administrator is managing the help desk system in terms of approving the submitted tickets, monitoring and updating the user'sticket status as well as providing proper solutions. The administrator can also upload knowledge-based solutions for instructional materials, inventory details and system settings. The administrator can generate a report which includes the ticket summary oftransactions and inventory list of IT equipment.

Activity Diagram

Windle and Abreo (2012) stated that activity diagram is use to visually model thedynamic behavior of a given part of the system. Another use of an activity diagram is to model use cases.

Figure 9 illustrates the Activity Diagram for Ticket Management

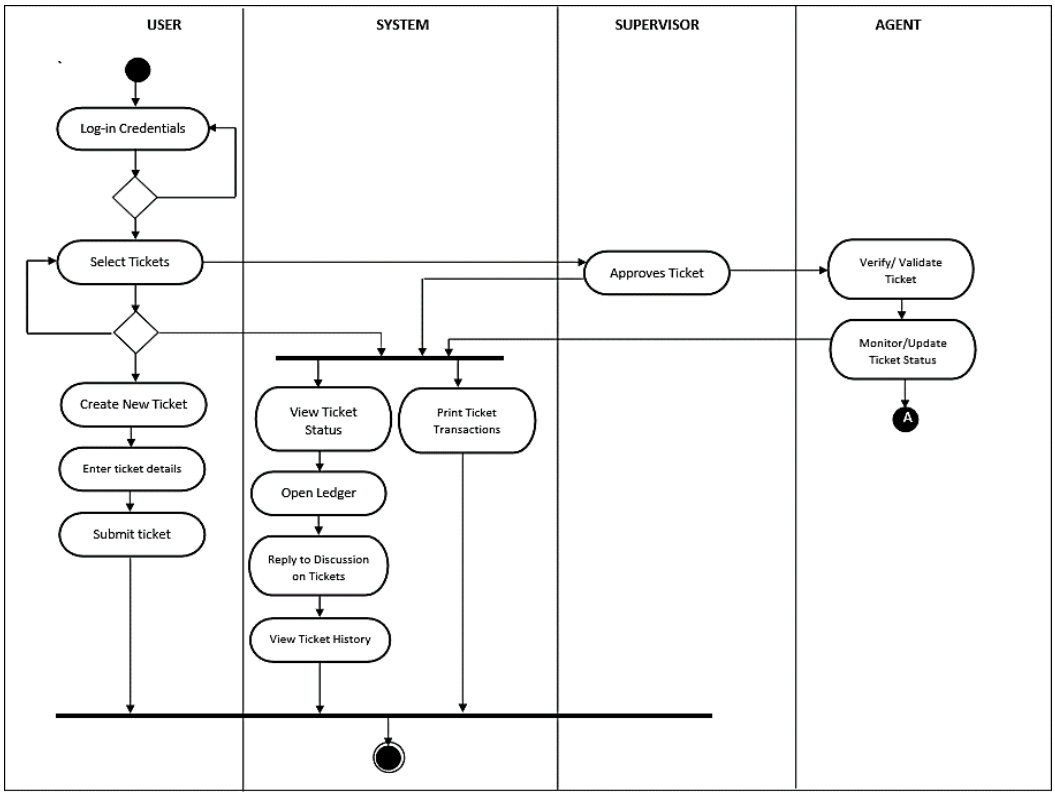


Figure 9. *System Activity Diagram for Ticket Management*

In the ticket management, the Dashboard requires a login access credential forsecurity and authenticity. Four different users of the system have their own level of restriction in using the system. Once access is granted, the user provides a different menu.

For ticket management, the User selects the Tickets menu to create a help desk ticket, then the details such as the type of the problem, the description and othersupplemental message are needed. The ticket number is then applied to the help desk ticket after submission. Once all the information is entered, then the ticket is submitted.The Supervisor can view all the submitted tickets and

approves each ticket based on priority or with the Date and Time received as shown in figure 9.

The Agent selects the same menu to verify the help desk tickets. The Agent canview the ticket Number for each and the Date and Time approved by the Supervisor.

The three users can view the ticket status, add and update the comments using theLedger, view the ticket history and print the ticket transactions.

Figure 10 illustrates the Activity Diagram for the Agent in Monitoring Ticket Status

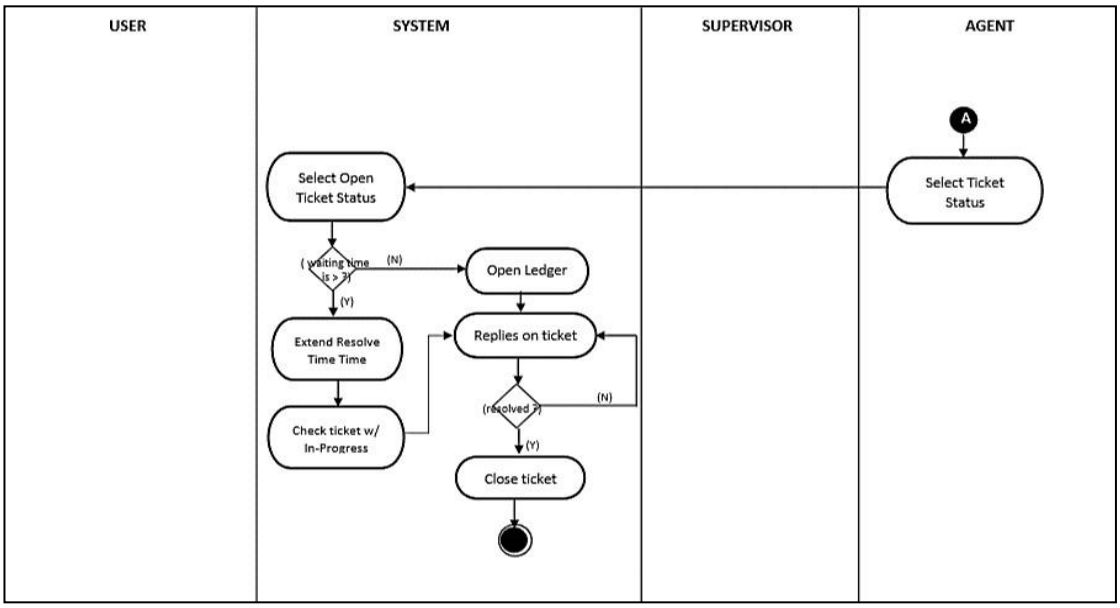


Figure 10. *System Activity Diagram for the Agent in Monitoring Ticket Status*

The diagram shows that within the Tickets menu, the Agent gets track of theticket status from Pending to Open and from in-Progress to Close.

The tickets Ledger option provides information for reviewing on the ticket status, priority, its ticket number, date and time requested and approved. Once the

Ledger is selected, the user can send a message or start a conversation thread with regards to theissue or problem requested.

The Agent first checks the Open tab for a queue of tickets with assigned priority; the name of the User including its designation and the office where it is assigned; and thewaiting time to solve the ticket request by the User. If the waiting time exceeds the targetresolution time for the ticket to resolve, then the Agent can extend the time for that particular ticket number; otherwise the Agent can verify the ticket and made a solution tothe problem. If the ticket has extended the resolution time, the ticket status is automatically forwarded to the In-Progress tab, which means the ticket is waiting for resolution to be made by the Agent. A resolution comment is added and updated throughthe use of the Ledger. Once the ticket is resolved, the Agent can already close the ticket.

Figure 11 illustrates the Activity Diagram for Procurement and InventoryManagement of Agent and Supervisor

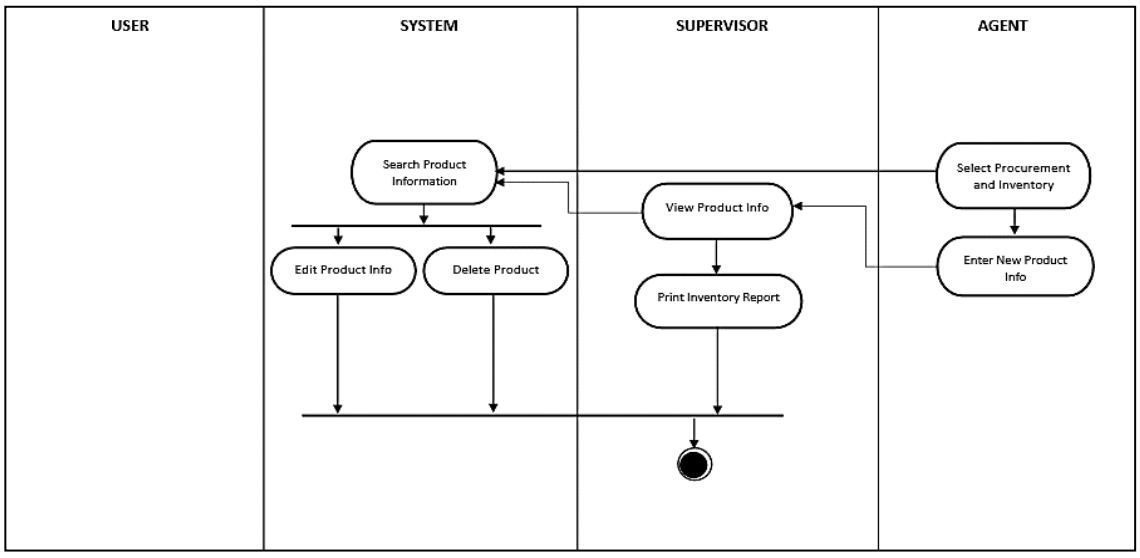


Figure 11. *System Activity Diagram for Procurement and Inventory*

Management ofAgent and Supervisor

In this process, the Supervisor and Agent can both access the Procurement andInventory Management. Once the Procurement and Inventory menu is selected, it provides a summary of the IT equipment of the office with its hardware and software specifications.

The Agent is responsible in adding and updating the list of products with the IT equipment inventory. The Supervisor is limited to view the list and generate a summary of the inventory through a report format. In addition, the Agent can make modifications of the detailed product information listed in the inventory such as search, edit and delete.

Figure 12 illustrates the Activity Diagram Management of Knowledge-BasedSolutions Panel

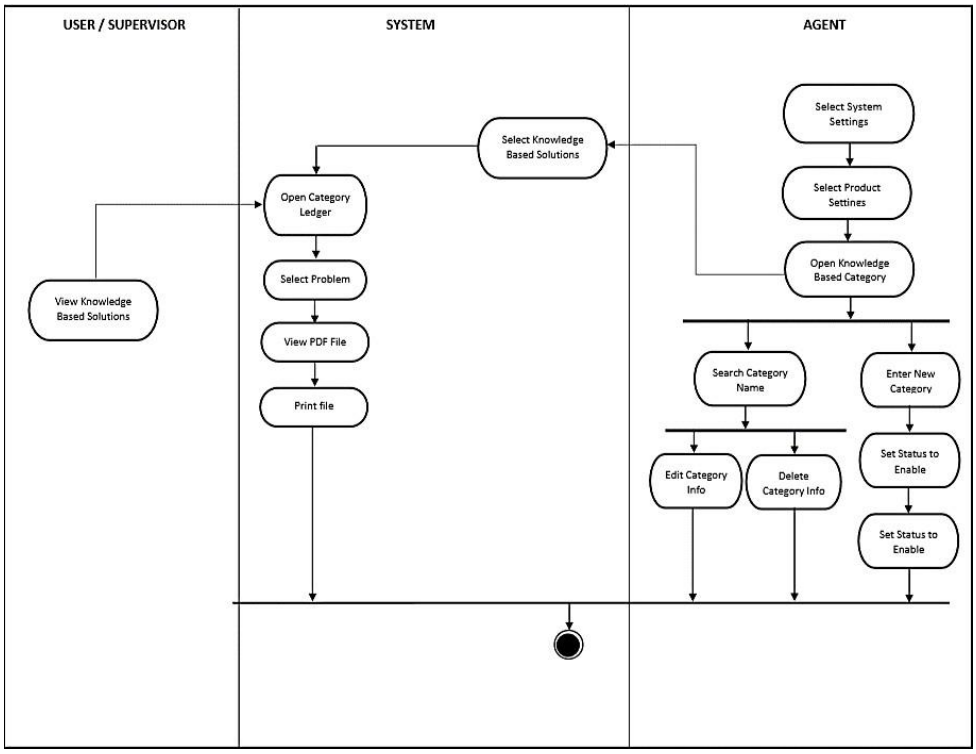


Figure 12. *Management of Knowledge-Based Solutions Panel*

As shown in the diagram, all users can access the knowledge base solutions menu. It provides the step by step guide or troubleshooting steps in solving most common IT related problems. It consists of different categories and is available in a PDF form whichcan be downloaded and printed by the users.

The Agent is granted access by adding and updating the different categories that is made available for other users. Once the Agent creates a new category and is set to Enable, a PDF file is uploaded to the system. Each category has a list of common problems that is frequently encountered by various group of users. Both the User and theSupervisor can view the files and have an option to download or print.

Figure 13 illustrates the Management of System Settings for Agent Activity Diagram.

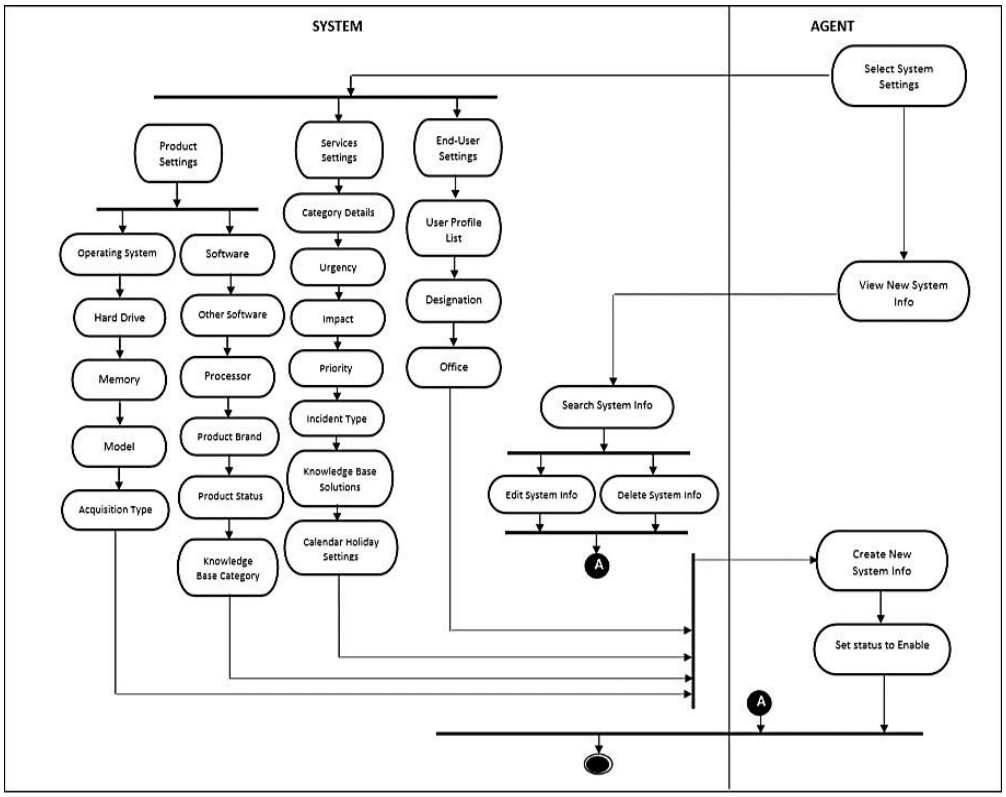


Figure 13. *Management of System Settings for Agent*

In managing the system settings, only the Agent and the Administrator can manage the System Settings and Inventory list. The System Settings are divided intothree sections, namely: product settings, services settings and end-user settings.

The product settings are linked to the Procurement and Inventory menu of the system which includes the specifications and description of IT equipment such as the operating system, hard drive, memory, product brand and model and acquisition type. The Services Settings are related to ticket management. This is where the Agent or Administrator sets the value for the urgency and impact of every help desk ticket created from High, Moderate to Low values. In this menu, the Agent can set the calendar date forthe Holidays. While the End User settings is managed for the user's profile, where the information about the employee is displayed such as last name, middle name, first name, address, contact number, email address, designation and division/section. This menu allows the Agent and the Administrator to update the record of the employees, set username and password and assign restriction for the type of users, namely: User, Agent,Supervisor and the Administrator.

Sequence Diagram

A sequence diagram shows object interactions arranged in time sequence. It depicts the objects and classes involved in the scenario and the sequence of messages exchanged between the objects needed to carry out the functionality of the scenario [43].

Figure 14 illustrates the Sequence Diagram for Log-in Access

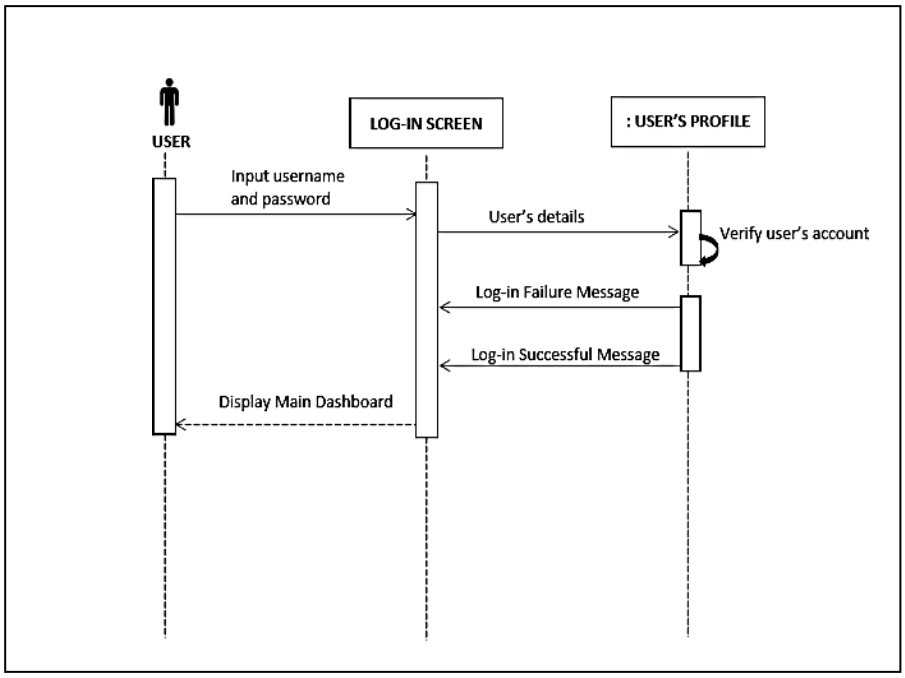


Figure 14. *Sequence Diagram for Log-in Access*

The diagram illustrates the sequence in giving access for the users in using thesystem. All users are required to have a username and password. The system log-in screen provides automatically the list of the username in a drop down manner in accordance to the type of access given.

Once the user's details are sent to the system, the account is verified along withthe user's profile and provides a response message if it is either a successful or a failedlog-in. If the user gets a successful log-in, it displays menus available for the user.

Figure 15 illustrates the Sequence Diagram for Tickets Activity.

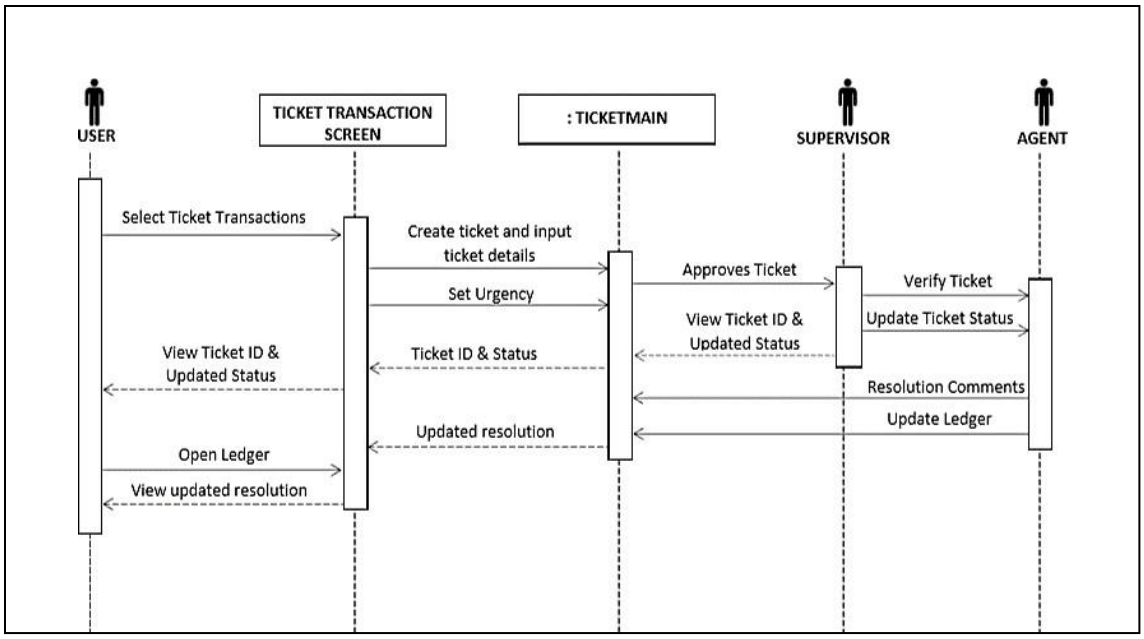


Figure 15. *Sequence Diagram for Tickets Activity*

The figure illustrates how the ticket is handled from creation to approval that isperformed by different users.

The user selects the Ticket Transaction screen from Tickets Menu. Tickets Ledgershows the list of ticket information; including the ticket status, priority, date and time request, date and time approved, request type and user's information. To create a ticket, the user requires information for the type of request, type of ICT equipment, and description of the problem. The user sets the value for urgency which is from low, medium to high. When the user submits the ticket, a ticket number is automatically created. The Supervisor approves the ticket from the Ticket Transaction Screen.

Unapproved incoming help desk tickets are queued in the "Pending" tab. Only theSupervisor can approve the tickets submitted by the User.

The Agent verifies ticket from the “Open” tab. Each ticket number has its own priority and waiting time. The Agent updates and keeps track the ticket transactions using the Ledger where it can add comment resolution and accessible to all other users. The User and Supervisor can view the ticket status, update and join with the discussion threadthrough the Ledger.

Figure 16 illustrates the Sequence Diagram of Monitoring Ticket Status

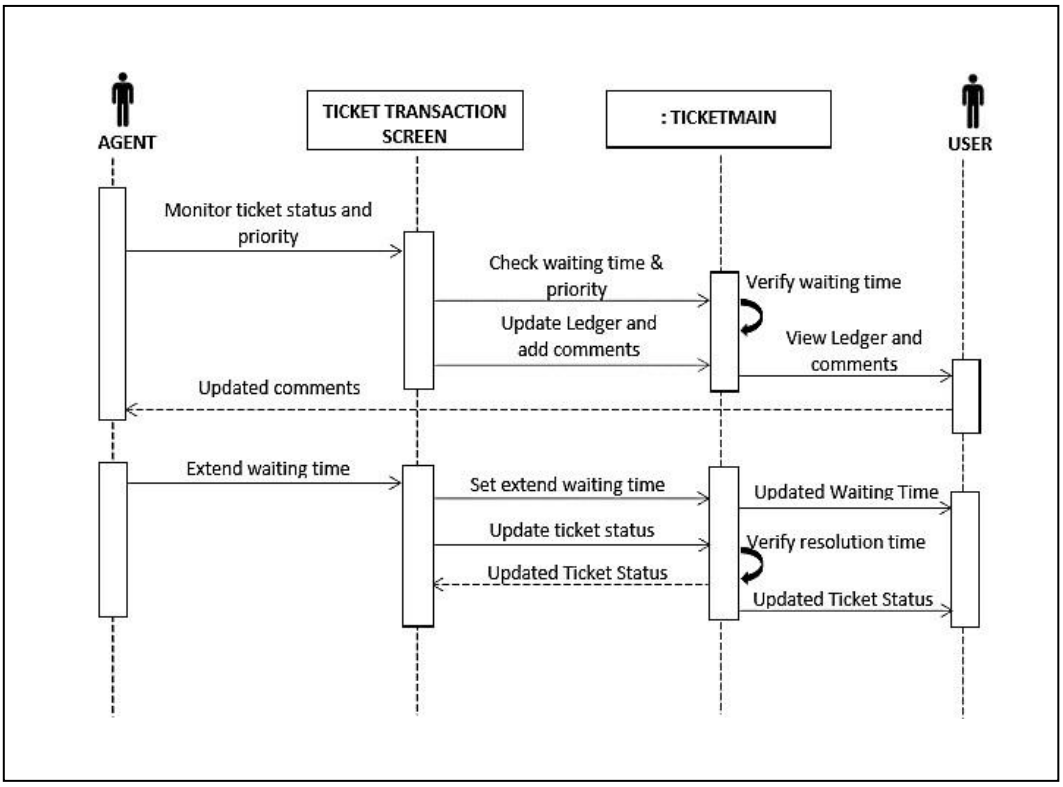


Figure 16. *Sequence Diagram of Monitoring Ticket Status*

In monitoring the ticket status, the Agent is responsible in monitoring the ticket status from the “Open” tab. The Agent verifies the waiting time and priority of each ticket. If the waiting time is within the target resolution time, the ticket is addressed and resolved. The ticket on a higher the priority is served first. But if the waiting time exceedsor nearly to exceed the target resolution time of the ticket being

served, then the Agent is allowed to extend the time. The system automatically computes the additional extensiontime for that particular ticket number.

If the ticket is extended, the ticket status is from “Open” to “In-Progress”. The Agent is only allowed to perform the time extension once. Once the resolution time haschanged due to time extension, the waiting time is also updated. Both the User and the Supervisor can view and track the movement of the tickets submitted and approved.

Figure 17 illustrates the Sequence Diagram for Procurement and InventoryManagement

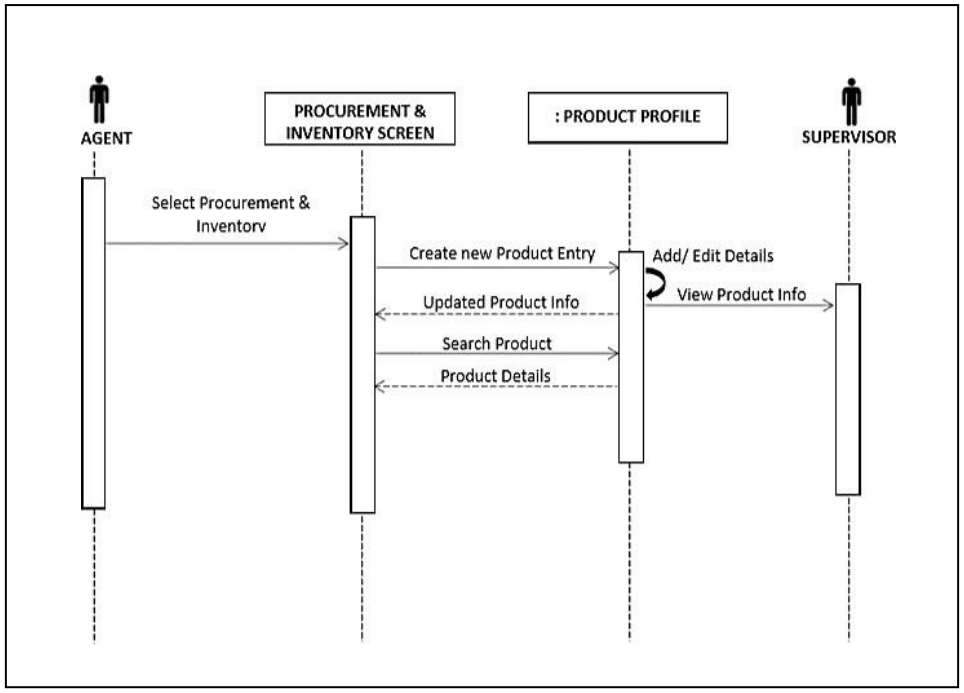


Figure 17. *Sequence Diagram for Procurement and Inventory Management*

The figure shows the Sequence Diagram for the procurement and inventory management. The Supervisor, Agent and Administrator have the access to this feature ofthe system. Only the Agent and the Administrator have the access to have a data entry on the different hardware and software specifications of an IT Equipment

such as the harddisk drive, memory, operating system installed and acquisition type. Once all the specifications are recorded, the Supervisor can view the ICT equipment with its specifications through the Procurement and Inventory section. Both users can perform various actions such as search and print an inventory report.

Requirement Analysis

The minimum hardware and software requirements are presented which wereused in the development of the system.

Table 1 shows the minimum hardware requirements of the system.

Table I

Hardware Specification

	Minimum Requirements
CPU Type Model	Dual Core 2.3 GHz or Higher
Storage Type	500 GB Hard Disk or Higher
Input	Mouse, Keyboard, Webcam
Output	Monitor, Printer

Table 2 shows the minimum software requirements for the system

Table II

Software Specification

	Minimum Requirements
Operating System	Microsoft Windows 7 or Higher
Programming Tool	Microsoft Access
Database	MySQL Server

System Architecture

The system architecture presents the physical view of the systems and shows theinterrelationship among different entities when implemented. Figure 18 illustrates the system architecture.

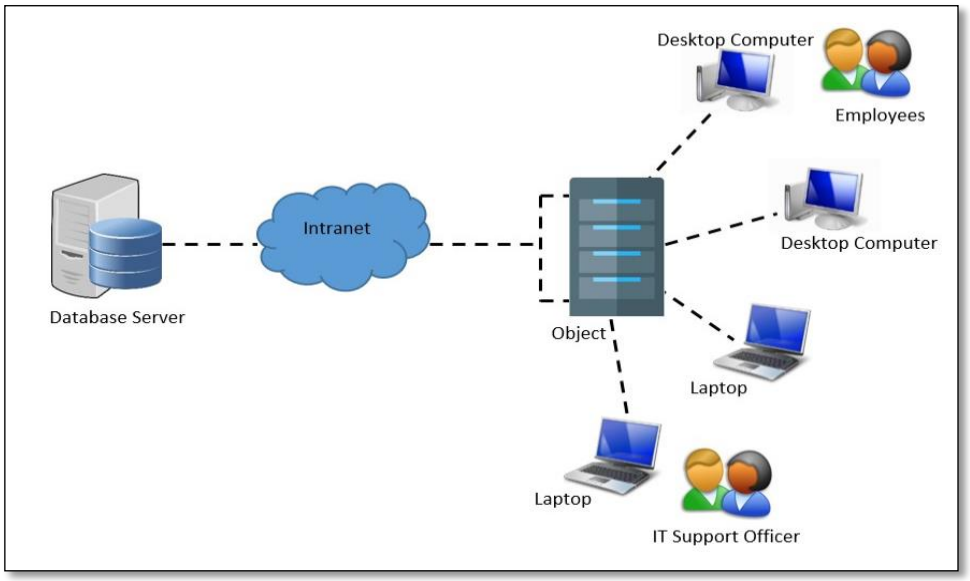


Figure 18. *System Architecture*

The figure shows the system architecture wherein it is accessible through a dashboard and works in an intranet technology where all desktops and computers areconnected. Employees and IT Support Officer as the users of the system can view thesystem as an object. The object is represented as a different component or integrated modules that are composed of a help desk management system such as the ticket generation for IT-related issues, IT asset inventory management, knowledge based solutions as a self-help tool and report management.

All employees and the IT Support Officer can access to the system through theirown dashboard using their user's account that has a different content of the system as tothe privileges assigned to them. Employees submit the IT related issues and inquiries wherein the IT Support Officer can receive and view submitted tickets with the corresponding ticket number. Help desk activities together with the records of IT assetsand repository of knowledge-based solutions are stored in the database server.

Chapter 4

Contents:

- Systems / Applications' Features and discussion

Note: Features must be arranged by objectives and in title case, and italicized.

Example:

Chapter 4

RESULTS AND DISCUSSION

This chapter includes the results and discussion of the systems' features: (a) create an account for teachers and principal, (b) design a platform for uploading of documents, (c) create a database for information storage, (d) manage information in e-portfolio, (e) view e-portfolio of the teachers, (f) download and print documents.

A. Create an Account for Teachers and Principal

Teachers can create their account on this online platform to access into the system. Figure 8 shows the interface of the sign-up form in creating an account.

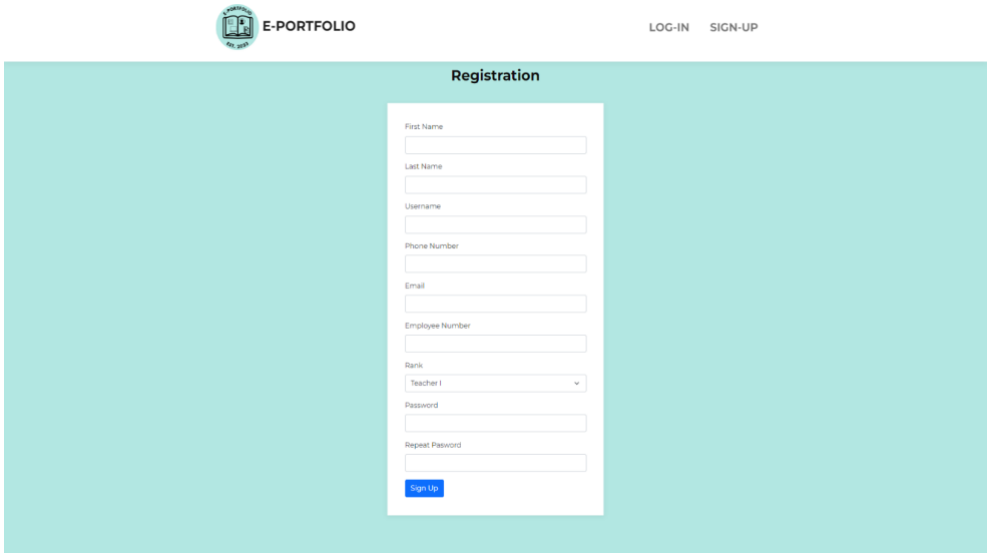


Figure 8. *Sign-Up Form*

To create an account, the teacher must fill out the necessary field with corresponding information needed such as their First Name, Last Name, Contact Number, Email Address, Username, and Password as shown in Figure 8.

Chapter 5

Contents:

- Findings
- Conclusions
- Recommendations

Note: Recommendations must be aligned to the beneficiaries of the study (Significance of the Study)

Example:

Chapter 5

SUMMARY, CONCLUSIONS, AND RECOMMENDATIONS

This chapter presents the summary of findings, conclusions, and recommendations of the study.

Summary

This study aimed to develop an online site for teacher's and principal a web-based system for E-Portfolio Web-Based System for Secondary School Teachers that manages the teachers' documents. The system has the capability to record the teachers and principal details for e-portfolio through the system's online site. It allows teachers to create account and upload their documents and submit it to the principal electronically with convenience and efficiency. With the documents availability online, teachers are able to use the system to manage the documents, view it as necessary, download and print it when needed with

ease and as quickly as possible.

The Feature Driven Development Model was used in the development of the system which is useful for the system's iterative approach on each phase of the E-Portfolio Web-Based System for Secondary School Teachers development.

The researchers used Hypertext Preprocessor (PHP) as the major software in developing the system. Structured Query Language (MySQL) as the database, Hypertext Markup Language (HTML) and Cascading Style Sheet (CSS) for design.

Conclusions

Based on the findings of the study, the E-Portfolio Web-Based System for Secondary School Teachers online portal was useful in can creating an account for the teachers and principal as it provided a viable platform where teachers' documents can be uploaded, can be managed, can be viewed, can be downloaded and printed as necessary with ease and in a speedy manner. However, it lacks capability to display the whole portfolio as a single file upon viewing which could disappoint the teachers and principal.

As to the results of this study, there is a need to upgrade the system and use a suitable programming language that will enhance its efficiency and effectiveness so it will be more responsive to the needs of the end - users.

Recommendations

Based on the preceding findings and conclusions, the following series of actions are recommended:

1. The teacher must be willing to learn and be trained on how to use the system during its implementation for it to be effective and useful for them.

2. The principal must implement the system in order to respond to the needs of the teachers in terms of preparing, managing and safekeeping of documents for E-Portfolio.
3. The future researchers must make use of the findings of this study in the development of other similar systems to further its effectiveness in the fields of technologies or industries.
4. The framework of the system may be updated to increase performance, effectiveness and fix system compatibility problems.
5. To produce a more refined output, the user interface design may be upgraded, to enhance the system.
6. The uploaded documents must be available anytime to be viewed by the teachers in order for them to update their works and as needed to be printed by the teacher and principal and have an updated information.
7. The system should be evaluated by the IT experts to ensure that it is as effective and efficient as possible before the system is actually integrated in school.

Sample references in APA format

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Sample Curriculum Vitae



RICHARD SANCHEZ

Product Designer

About Me

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Vestibulum sit amet quam rhoncus, egestas dui eget, malesuada justo. Ut aliquam augue.

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123 Anywhere St., Any City

LANGUAGE

- English
- Germany (basic)
- Spain (basic)

EXPERTISE

- Management Skills
- Creativity
- Digital Marketing
- Negotiation
- Critical Thinking
- Leadership

EXPERIENCE

Studio Showde
Canberra - Australia
2020 - 2022
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Kota Baru - Singapore
2016 - 2020
Lorem ipsum dolor sit amet, consectetur adipiscing elit. Vestibulum sit amet quam rhoncus, egestas dui eget, malesuada justo. Ut aliquam augue.

Studio Showde
sydney - Australia
2010 - 2015
Lorem ipsum dolor sit amet, consectetur adipiscing elit. Vestibulum sit amet quam rhoncus, egestas dui eget, malesuada justo. Ut aliquam augue.

EDUCATION

Borcelle University
Bachelor of Business Management
2014-2023

Borcelle University
Master of Business Management
2014-2018

SKILLS SUMMARY

Design Process 78 %

Project Management 81 %